Dennis Frauen

Ferdinand-Miller-Platz 10
80335 Munich
+4915142074707
frauen@lmu.de
Date of birth: 28.11.1997

About

I am a PhD student and researcher at the Institute of AI in Management, LMU Munich. My research focuses mainly on causal machine learning. More precisely, I am interested in developing novel methodology for causal inference from observational data. In doing so, I build on tools from both statistical theory as well as recent advances in e.g., Deep Learning.

EDUCATION

Sep.2021 – today
LMU Munich PhD Computer Science
- Supervisor: Prof. Dr. Stefan Feuerriegel

Oct. 2019 – Aug. 2021
University of Göttingen MSc Mathematics
- GPA: 1.1 ("Very good", german grading scale ranging from 1.0 to 5.0)
- Focus on mathematical statistics with minor courses in applied statistics, machine learning and applied mathematics
- Master thesis on statistical inference based on optimal transport for protein alignment (Supervisors: Axel Munk, Housen Li)

Aug. 2018 - Jan. 2019
Lund University Erasmus Exchange Semester in Mathematics
- Passed with Distinction

Sep. 2016 - Sep. 2019
University of Hamburg BSc Mathematics
- GPA: 1.36 ("Very good", german grading scale ranging from 1.0 to 5.0)
- General studies in pure and applied Mathematics
- Minor subject: Physics

Aug. 2016
Otto-Hahn-Gymnasium Geesthacht Abitur (high school graduation)

PUBLICATIONS

Melnychuk, V., Frauen, D., & Feuerriegel, S. (2022). Causal transformer for estimating counterfactual outcomes. (ICML 2022)

Frauen, D., Hatt, T., Melnychuk, V., & Feuerriegel, S. (2022). Estimating average causal effects from patient trajectories. *arXiv preprint arXiv:2203.01228*. (Under review)

Frauen, D., Melnychuk, V., & Feuerriegel, S. (2022). Estimating individual treatment effects under unobserved confounding using binary instruments. arXiv preprint *arXiv:2208.08544.* (Under review)

Melnychuk, V., Frauen, D., & Feuerriegel, S. (2022). Normalizing flows for interventional density estimation. *arXiv preprint arXiv:2209.06203.* (Under review)

WORK & RESEARCH EXPERIENCE

Nov. 2020 – Aug. 2021
Teaching Assistant, University of Göttingen, Department of Mathematics
- Tutor for the lectures "Statistical foundations of data science 2" (Master) and "Discrete stochastics" (Bachelor)

Jul. 2020 - Sep. 2020 **Summer Research Internship (Remote), ETH Zurich, Department of Computer Science**

- Remote Research (due to COVID-19) under supervision of Fanny Yang (Department of Computer Science) and Armeen Taeb (Seminar of Statistics)
- Responsible for algorithm implementation and performing simulation experiments in R
- Weekly participation in the group meetings of the Statistical Machine Learning group and presentation of results

Feb. 2019 - Mar. 2019 **Data Science Internship, Hauni Maschinenbau GmbH, Hamburg**

- Statistical data analysis, data pre-processing and presentation of results

Awards

Oct. 2019 - Sep. 2021 **German excellence scholarship (Deutschlandstipendium)**

- Scholarship holder of the Germany Scholarship for academic achievements

SKILLS & INTERESTS

Languages	English (fluent), German(native), Spanish (CEFR A1)
Programming	Python, R, SQL, LaTeX, MATLAB, Java, Git
Python Packages	PyTorch (lightning), TensorFlow + Keras, Pandas, NumPy, SciPy, Pandas, matplotlib
Volunteering	Member of ESN Munich, welcoming international students to the university
Other Interests	Piano and bass playing, table tennis, squash